

## PROGRAM 2

### Add and multiply two matrices

```
#include <stdio.h>

int main() {

    int m, n, p, q;

    int i, j, k, choice;


    // Input dimensions of first matrix

    printf("Enter rows and columns of first matrix: ");

    scanf("%d %d", &m, &n);


    int A[m][n];

    printf("Enter elements of first matrix:\n");

    for (i = 0; i < m; i++) {

        for (j = 0; j < n; j++) {

            scanf("%d", &A[i][j]);

        }

    }


    // Input dimensions of second matrix

    printf("Enter rows and columns of second matrix: ");

    scanf("%d %d", &p, &q);
```

```
int B[p][q];  
printf("Enter elements of second matrix:\n");  
for (i = 0; i < p; i++) {  
    for (j = 0; j < q; j++) {  
        scanf("%d", &B[i][j]);  
    }  
}
```

```
// Display matrices  
printf("\nFirst Matrix:\n");  
for (i = 0; i < m; i++) {  
    for (j = 0; j < n; j++) {  
        printf("%d ", A[i][j]);  
    }  
    printf("\n");  
}
```

```
printf("\nSecond Matrix:\n");  
for (i = 0; i < p; i++) {  
    for (j = 0; j < q; j++) {  
        printf("%d ", B[i][j]);  
    }  
    printf("\n");  
}
```

```
}
```

```
// Menu
```

```
printf("\nChoose an operation:\n");
```

```
printf("1. Add Matrices\n");
```

```
printf("2. Multiply Matrices\n");
```

```
printf("Enter choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice) {
```

```
    case 1:
```

```
        if (m == p && n == q) {
```

```
            int Sum[m][n];
```

```
            printf("\nResult of Addition:\n");
```

```
            for (i = 0; i < m; i++) {
```

```
                for (j = 0; j < n; j++) {
```

```
                    Sum[i][j] = A[i][j] + B[i][j];
```

```
                    printf("%d ", Sum[i][j]);
```

```
                }
```

```
                printf("\n");
```

```
            }
```

```
        } else {
```

```
            printf("\nMatrix addition not possible (dimensions must match).\n");
```

```
}
```

```
break;
```

case 2:

```
if (n == p) {
```

```
    int Prod[m][q];
```

```
    printf("\nResult of Multiplication:\n");
```

```
    for (i = 0; i < m; i++) {
```

```
        for (j = 0; j < q; j++) {
```

```
            Prod[i][j] = 0;
```

```
            for (k = 0; k < n; k++) {
```

```
                Prod[i][j] += A[i][k] * B[k][j];
```

```
            }
```

```
            printf("%d ", Prod[i][j]);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
} else {
```

```
    printf("\nMatrix multiplication not possible (columns of first must equal  
rows of second).\n");
```

```
}
```

```
break;
```

default:

```
printf("\nInvalid choice.\n");
```

```
}
```

```
return 0;
```

```
}
```